

An illustration of a person with dark hair, wearing a dark hoodie, sitting at a desk and typing on a keyboard. They are looking at three large computer monitors displaying code. The desk is cluttered with a pen holder, a mug, and some papers. In the background, there are potted plants and a window with a brick wall.

Expert-Level Full Stack Developer Hiring Challenge

VendorFlow: A Production-Grade SaaS Platform Assessment for Decodio Softwares

Challenge Overview

Objective

Evaluate comprehensive full stack engineering capabilities through building a production-grade B2B procurement platform called **VendorFlow**. This assessment tests system design, backend and frontend engineering, database optimization, security, DevOps, and real-world problem-solving skills.

Target Platform

VendorFlow is a B2B procurement and vendor management platform used by medium-sized companies to manage vendors, create purchase requests, approve workflows, generate invoices, track inventory, handle multi-role access, and audit every action.



System Design

Architecture and scalability planning



Backend Engineering

Complex business logic implementation



Security

Threat modeling and mitigation



DevOps

Deployment and monitoring

Technical Stack Requirements

Candidate may choose any modern stack but must justify every architectural decision with clear reasoning.



Frontend Options

- Next.js
- React
- Vue
- Svelte



Backend Options

- Node.js / NestJS / Express
- Django / FastAPI
- Spring Boot
- ASP.NET



Database

PostgreSQL (required)

Redis for caching



Infrastructure

Docker mandatory

Infrastructure choice up to candidate

Hard Rules

01

No Full CRUD Demo Projects

Simple CRUD applications result in automatic failure. System must demonstrate complex business logic, event handling, authorization hierarchy, async processing, and transaction safety.

02

Live Architecture Defense

Candidate must explain architectural decisions, database schema choices, tradeoffs, scaling strategy, security considerations, and failure handling during review.

03

Midway Requirement Change

Interviewer introduces requirement changes during review. Candidate must adapt architecture live. This is the strongest anti-AI evaluation layer.

04

Git History Evaluation

Meaningful commits, progressive implementation, feature branches, and detailed pull request descriptions required. Single giant commits penalized.

05

AI Usage Disclosure

Document which tools were used, which sections were AI-assisted, and what modifications were personally made. Undisclosed heavy AI dependence results in rejection.

Core Features: Part 1



Multi-Tenant Architecture

Support multiple companies with isolated data access, separate users, vendors, inventory, and procurement workflows. Tenant-aware database design with middleware-based tenant resolution and secure isolation preventing cross-tenant leakage.



Procurement Workflow Engine

Configurable approval engine with dynamic workflow stages, conditional approval logic, amount-based approvals, escalation rules, timeout handling, parallel approvals, rejection rollback, and audit logging. Workflow rules must be configurable from UI without hardcoded workflows.



Authentication & Authorization

JWT authentication with refresh tokens, session invalidation, role-based and permission-based access control, temporary access delegation, and MFA support. Roles include Super Admin, Company Admin, Procurement Officer, Finance Officer, Inventory Manager, Auditor, and Read-Only Viewer.



Real-Time Collaboration

Live approval updates, notifications, presence system, concurrent editing protection, and optimistic UI updates. Candidate must justify WebSockets vs SSE, conflict resolution strategy, and offline handling strategy.

Core Features: Part 2

Vendor Management

Vendor onboarding, scoring system, compliance document uploads, expiration alerts, blacklisting, risk scoring, and performance metrics. Uploaded files must be securely handled.

Inventory Tracking

Inventory movement logs, warehouse support, reservations, purchase-linked stock updates, batch operations, and reconciliation. Prevent race conditions using transactions, locking, isolation levels, and concurrency handling.

Audit Logging System

Log every important action with before/after values, actor, IP address, timestamp, tenant, and device/session. Immutable audit records with search/filter support, pagination, and export capabilities.

Analytics Dashboard

Dashboards for procurement trends, vendor performance, approval bottlenecks, inventory movement, and user activity. Analytics queries must be optimized with clear indexing strategy, materialized views, and caching explanations.

Core Features: Part 3



API Design

Versioned APIs with OpenAPI/Swagger docs, rate limiting, input validation, pagination, filtering, sorting, idempotency, proper HTTP semantics, and error handling standards. Include API request tracing, correlation IDs, and structured logging.



Background Jobs

Use queues for email sending, notifications, invoice generation, scheduled escalation, vendor compliance reminders, and report generation. Explain retry strategy, dead-letter handling, idempotency, and queue scaling.



Frontend Architecture

Production-grade architecture with component modularity, state management, form handling, optimistic updates, error boundaries, accessibility, responsive design, data caching, lazy loading, and performance optimization.

A stylized illustration of a man with dark hair, wearing a grey jacket, standing in a server room. He is interacting with a futuristic, glowing blue interface that appears to be a transparent screen or a set of data visualizations. The background shows rows of server racks with glowing lights and cables, creating a high-tech, digital environment.

Security & DevOps Requirements

Security Requirements

Candidate must proactively address:

- SQL injection
- XSS and CSRF
- Broken access control
- JWT vulnerabilities
- File upload attacks
- Rate limiting
- Password security
- Secrets management

⚠️ Mandatory: Submit comprehensive Threat Model Document including attack vectors, mitigations, assumptions, and risk severity assessment.

DevOps & Deployment

Mandatory requirements:

- Dockerized services with Docker Compose
- Environment separation
- CI/CD pipeline
- Health checks
- Structured logging
- Monitoring setup
- Error tracking strategy

Provide deployment guide, environment configuration, and rollback strategy. Kubernetes deployment is a bonus.

Performance & Testing Requirements

100K+

Users Supported

System must handle 100,000+ users with proper horizontal scaling strategy

10K

Concurrent Approvals

Support 10,000 concurrent approvals with optimized database queries

100%

Test Coverage

Critical workflows, authorization, concurrency, and failure scenarios

Testing Requirements

- Unit Tests

Core business logic and utility functions

- API Tests

Endpoint validation and error handling

- Integration Tests

Service interactions and database operations

- End-to-End Tests

Complete user workflows and scenarios

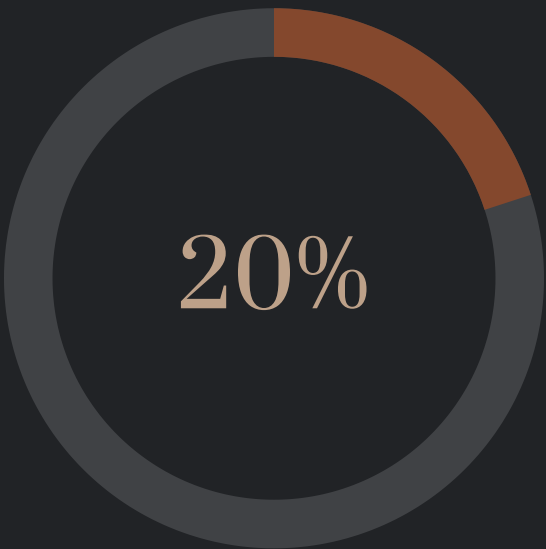
Candidate must explain bottleneck identification, horizontal scaling, database partitioning, cache invalidation, CDN usage, and read replicas strategy.

Deliverables & Evaluation

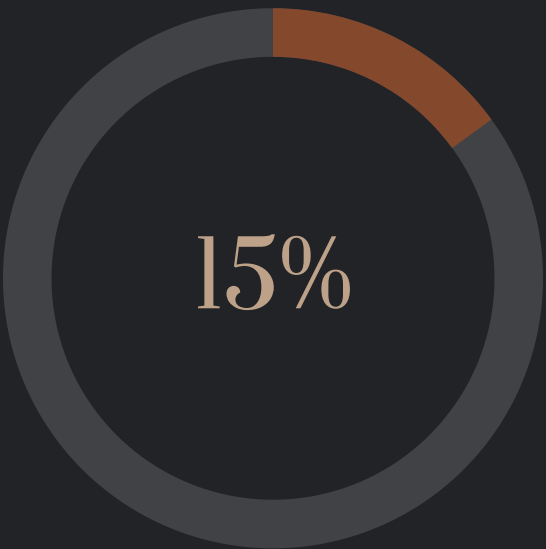
Required Deliverables

01	02	03
Source Code	README	System Design Document
Complete Git repository with meaningful commits and feature branches	Setup instructions, architecture overview, tradeoffs, assumptions, security and scaling considerations	High-level architecture, ERD diagrams, service communication, caching and queue strategies, scaling plan, security model
04	05	
API Documentation	Threat Model	
Swagger/OpenAPI with example requests and authentication examples	Comprehensive security analysis with attack vectors and mitigations	

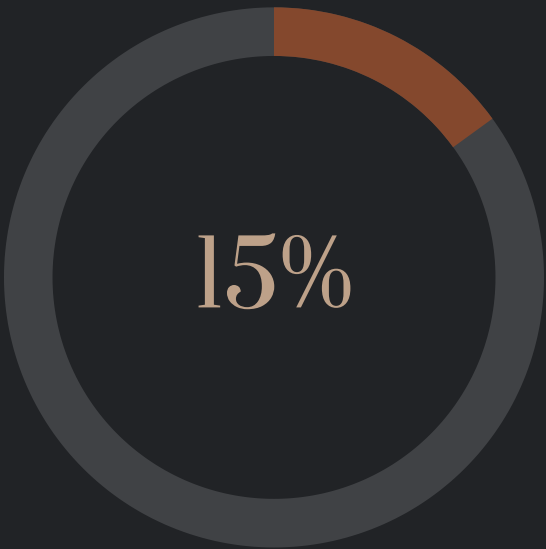
Evaluation Rubric



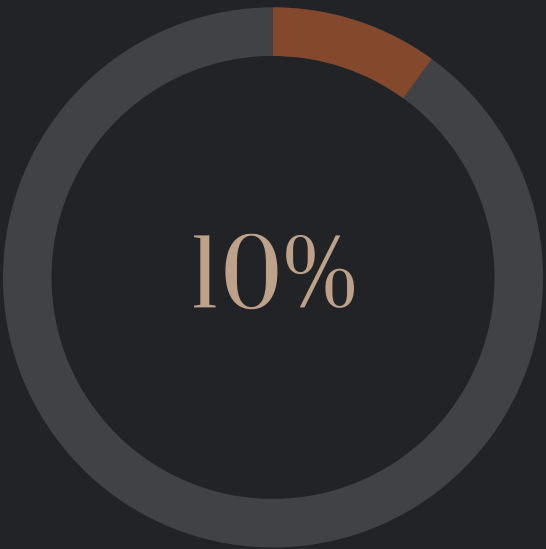
System Design



Backend Engineering



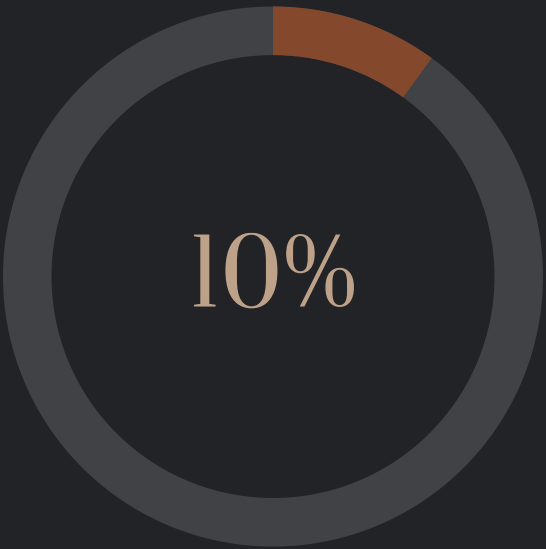
Security



Scalability Thinking



Database Engineering



Frontend Engineering

✔ **Time Expectation:** Senior Engineer: 20-35 hours | Lead Engineer: 15-25 hours. Evaluation prioritizes architecture quality, decision making, engineering depth, code quality, scalability reasoning, and security maturity over feature completion.